

KPI Threshold Table

KPI	Good	Warning	Critical
RSRP	>-90 dBm	-90 to -105	<-105
SINR	>20 dB	10-20	<10
Throughput	>100 Mbps	50-100	<50
Latency	<20 ms	20-50	>50
Packet Loss	<1%	1-3%	>3%

- 1) **RSRP (Reference Signal Received Power):** Measures the strength of the received signal from the base station.
 - Good (>-90 dBm): Strong signal with excellent coverage.
 - Warning (-90 to -105 dBm): Moderate signal strength, users may experience reduced performance
 - Critical (<-105 dBm): Weak signal causing poor connectivity, call drops and slow internet.
- 2) **SINR (Signal to Interference plus Noise Ratio):** Measures the quality of the received signal by comparing the useful signal against interference and noise.
 - Good (>20dB): Excellent signal quality with minimal interference.
 - Warning (10-20 dB): Moderate interference that may reduce throughput.
 - Critical (<10dB): High interference leading to poor network performance.
- 3) **Throughput:** Represents the actual data transfer rate experienced by the user.

- Good (>100 Mbps): High-speed data transfer suitable for streaming, gaming, and large downloads.
- Warning (50-100 Mbps): Acceptable performance with slight reduction in speed.
- Critical (<50 Mbps): Slow internet speed affecting user experience.

4) **Latency:** Measures the time required for data to travel between the user and the network.

- Good (<20 ms): Very responsive network suitable for real-time applications.
- Warning (20-50 ms): Slight delay that may affect online gaming and video calls.
- Critical (> 50 ms): High delay resulting in buffering and poor application performance.

5) **Packet Loss:** Represents the percentage of data packets that fail to reach their destination.

- Good (<1%): Reliable communication with negligible data loss
- Warning (1-3%): Minor packet loss that may impact voice and video quality.
- Critical (>3%): Significant packet loss causing interruptions, retransmissions, and degraded network performance.